23AD31SC–FULL STACK DEVELOPMENT-II

Course Category:	Skill Enhancement course	**Credits:**	2
Course Type:	Theory	**Lecture-Tutorial-Practical:**	0-1-2
Prerequisite:	• Knowledge in Full Stack Development	**Sessional Evaluation:** **Univ. Exam Evaluation:** **Total Marks:**	30 70 100
Course Objectives:	**Students undergoing this course are expected:**		
	• Make use of Modern- day JavaScript with ES6 standards for designing Dynamic web pages • Building robust & responsive User Interfaces using popular JavaScript library _React.js‘. Building robust backend APIs using _Express. js‘ • Establishing the connection between frontend (React) User interfaces and backend APIs (Express) with Data Bases(My SQL) • Familiarize students with GitHub for remote repository hosting and collaborative development.		

Course Outcomes:	**Upon successful completion of the course, the students will be able to:**	
	CO1	Building fast and interactive UIs
	CO2	Applying Declarative approach for developing web apps
	CO3	Understanding ES6 features to embrace modern JavaScript
	CO4	Building reliable APIs with Express. Js
	CO5	Create and manage Git repositories, track changes, and push code to GitHub.
Course Content:	**Experiments covering the Topics:** • Introduction to DOM (Document Object Model), Ecma Script (ES6) standards and features like Arrow functions, Spread operator, Rest operator, Type coercion, Type hoisting, String literals, Array and Object Destructuring. • Basics of React. js like React Components, JSX, Conditional rendering Differences between Real DOM and Virtual DOM. • Important React.js concepts like React hooks, Props, React forms, Fetch API, Iterative rendering using JavaScript map() function. • JavaScript runtime environment node. js and its uses, Express. js and Routing, Micro-Services architecture and MVC architecture, database connectivity using (My SQL) • Introduction to My SQL, setting up MySQL and configuring, Databases, My SQL queries, subqueries, creating My SQL driver for database connectivity to Express. js server. • Introduction to Git and GitHub and upload project& team collaboration	

Sample Experiments:

1. **Introduction to Modern JavaScript and DOM**
   - **a.** Write a JavaScript program to link JavaScript file with the HTML page
   - **b.** Write a JavaScript program to select the elements in HTML page using selectors
   - **c.** Write a JavaScript program to implement the event listeners
   - **d.** Write a JavaScript program to handle the click events for the HTML button elements
   - **e.** Write a JavaScript program to With three types of functions
     - **i.** Function declaration
     - **ii.** Function definition
     - **iii.** Arrow functions
2. **Basics of React. js**
   - **a.** Write a React program to implement a counter button using react class components
   - **b.** Write a React program to implement a counter button using react functional components
   - **c.** Write a React program to handle the button click events in functional component
   - **d.** Write a React program to conditionally render a component in the browser
   - **e.** Write a React program to display text using String literals
3. **Important concepts of React. js**
   - **f.** Write a React program to implement a counter button using React use State hook
   - **g.** Write a React program to fetch the data from an API using React use Effect hook
   - **h.** Write a React program with two react components sharing data using Props.
   - **i.** Write a React program to implement the forms in react
   - **j.** Write a React program to implement the iterative rendering using map() function.
4. **Introduction to Git and GitHub**

k. **Setup**
- Install Git on local machine.
- Configure Git (user name, email).
- Create GitHub account and generate a personal access token.

l. **Basic Git Workflow**
- Create a local repository using git init
- Create and add files → git add .
- Commit files → git commit -m "Initial commit"
- Connect to GitHub remote → git remote add origin <repo_url>
- Push to GitHub → git push -u origin main

m. **Branching and Collaboration**
- Create a branch → git checkout -b feature1
- Merge branch to main → git merge feature1
- Resolve merge conflicts (guided)

	5. Upload React Project to GitHub ○ Create a new React app using npx create-react-app myapp ○ Initialize a git repo and push to GitHub ○ Use .gitignore to exclude node_modules ○ Create multiple branches: feature/navbar, feature/form ○ Practice merge and pull requests (can use GitHub GUI) **3. Introduction to Node. js and Express. js** **a.** Write a program to implement the _hello world‘ message in the route through the browser using Express **b.** Write a program to develop a small website with multiple routes using Express. js **c.** Write a program to print the _hello world‘ in the browser console using Express. js **d.** Write a program to implement the CRUD operations using Express. js **e.** Write a program to establish the Connection between A PIand Database using Express – My SQL drive **7. Introduction to My SQL** a. Write a program to create a Database and table inside that database using My SQL Command line client b. Write a My SQL queries to create table, and insert the data, update the data in the table c. Write a My SQL queries to implement the subqueries in the My SQL commandline client d. Write a My SQL program to create the script files in the My SQL workbench e. Write a My SQL program to create a database directory in Project and initialize a database. sql file to integrate the database into API **8. Team Collaboration Using GitHub** • Form groups of 2–3 students • reate a shared GitHub repo • Assign tasks and work in branches • Use Issues, Pull Requests, and Code Reviews Document code with README.md
Text Books & References Books:	**TEXTBOOKS:** 1. Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett Professional JavaScript for Web Developers Book by Nicholas C. Zakas 2. John Dean, Web Programming with HTML5, CSS and JavaScript, Jones & Bartlett Learning, 2019. 3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O‘Reilly. 4. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by-Step Guide to Creating Dynamic Websites by Robin Nixon 5. AZAT MARDAN, Full Stack Java Script: Learn Back bone. js, Node.jsand Mongo DB.2015

	REFERENCE BOOKS: 1. Full-Stack JavaScript Development by Eric Bush. 2. Programming the World Wide Web, 7th Edition, Robet W Sebesta, Pearson, 2013. 3. Tomasz Dyl ,KamilPrzeorski , MaciejCzarnecki, Mastering Full Stack React Web Development 2017
Online Learning Resources:	1. https://ict.iitk.ac.in/product/full-stack-developer-html5-css3-js-bootstrap-php-4/ 2. https://www.w3schools.com/html 3. https://www.w3schools.com/css 4. https://www.w3schools.com/js/ 5. https://www.w3schools.com/nodejs 6. https://www.w3schools.com/typescript 7. https://docs.github.com/ 8. https://education.github.com/git-cheat-sheet-education.pdf